Data Analysis of Electron Channel of the Drell–Yan Process using CMS open data

DY Signal and Background Selection Study

Khushpreet Kaur
M.Sc (Physics), Panjab
Chandigarh, India
k.kaur.pu@gmail.com

April 16, 2026

Abstract

This study presents a detailed cut-flow analysis of the Drell–Yan ($q\bar{q} \to Z/\gamma^* \to e^+e^-$) process in the di-electron final state using Monte Carlo samples corresponding to $\sqrt{s} = 13$ TeV proton-proton collisions. The primary objective is to isolate a high-purity signal sample by suppressing dominant Standard Model backgrounds, including $t\bar{t}$, single top, and diboson processes.

A sequential event selection strategy is implemented, beginning with a strict pre-selection of two oppositely charged, tightly identified electrons, followed by optimized kinematic and topological cuts. Key discriminating variables include the dilepton invariant mass, missing transverse energy (MET), transverse momentum of the dilepton system, and azimuthal angular separation.

The analysis demonstrates that the invariant mass window effectively isolates the resonant Z boson contribution, while MET and topological constraints significantly suppress neutrino-rich and multi-body background processes. The final selection achieves a high signal purity exceeding 90% while maintaining a controlled signal efficiency.

This work establishes a clean $Z \to e^+e^-$ control region and demonstrates the effectiveness of CMS-style selection strategies for precision electroweak studies and future beyond-the-Standard-Model searches in the dilepton channel.

Key Results: The full selection reduces the pre-selected sample of 227,864 events to 72,464 events, with 67,124 attributed to the Drell–Yan signal and 5,340 to residual backgrounds. This corresponds to a signal purity of 92.63% and a signal-to-background ratio of $S/B \approx 12.57$. The signal efficiency is 4.68%, while background rejection exceeds 90% for dominant processes such as $t\bar{t}$ and single top production. The resulting statistical significance, $S/\sqrt{S+B} \approx 248$, confirms a highly clean and signal-dominated dataset, establishing a robust $Z \to e^+e^-$ control region for precision studies.

Contents

1 Introduction **3**
1.1 Drell Yan Process . . . 3
1.2 The Large Hadron Collider (LHC) . . . 4
1.3 The Compact Muon Solenoid (CMS) Detector . . . 5
1.4 Electron Reconstruction and Identification . . . 5
1.5 Focus of Present Study . . . 5
1.6 Physics Motivation . . . 6
1.7 Analysis Goals . . . 6

2 Signal and Backgrounds Characterization **6**
2.1 Drell-Yan Process . . . 6
2.2 Background Processes . . . 6

3 Monte Carlo Samples **7**

4 Datasets and Simulated Samples **8**

5 Kinematic Observables and Selection Cuts **8**
5.1 Di-electron Invariant Mass (m_{ee}) . . . 8

6 Event Selection and Optimization Strategy **9**
6.1 Pre-Selection Criteria . . . 9
6.2 Kinematic Distributions of the Drell-Yan Signal . . . 10
6.3 Final Kinematic Cuts . . . 10

7 Results and Yields **13**
7.1 Discussion of Background Suppression Mechanisms . . . 14
7.2 Comparison with Standard Model Predictions . . . 15

8 Conclusion and Future Scope **15**
8.1 Future Scope . . . 15
.1 Data Sets . . . 16
.2 Code and analysis strategy . . . 17

1 Introduction

The study of the Drell-Yan (DY) process at the Compact Muon Solenoid (CMS) experiment is essential for understanding fundamental particle interactions at the Large Hadron Collider (LHC) [13, 14]. In this study, we focus specifically on the DY process in the electron-positron (e^+e^-) also referred as di-electron final state, where a quark and an antiquark annihilate to produce a Z boson, which subsequently decays into an electron-positron pair. This process constitutes a fundamental aspect of the Standard Model (SM) [7], and a thorough understanding of it is essential in high-energy physics as well as in the pursuit of new physics. It serves as a critical benchmark for investigating high-energy collisions and for testing the validity of theoretical models.

Although the DY process has high cross-section, it is accompanied by various background processes that can mimic the electron-positron final state. These background events must be carefully identified and mitigated to ensure a clean signal extraction. The presence of top-quarks pair production, single top quark production, and di-boson events complicates the extraction of the pure signal [18, 20]. Therefore, improving the signal purity is necessary to obtain more accurate measurements.

1.1 Drell Yan Process

The Drell-Yan process is a fundamental mechanism in high-energy particle physics, describing lepton pair production in hadron-hadron collisions. It was first suggested by Sidney D. Drell and Tung-Mow Yan in 1970 [12].

In this process, a quark from one hadron and an antiquark from another annihilate into a virtual photon or a Z boson, which then decays into a lepton pair as shown in the schematic:

$$q\bar{q} \longrightarrow Z \longrightarrow e^+e^- \tag{1}$$

Figure 1: Feynman diagram of the Drell-Yan process: quark-antiquark annihilation producing a virtual photon or Z boson, which decays into an electron-positron pair.

As the invariant mass of the lepton pair ($M_{\ell\ell}$) approaches the mass of the Z boson (M_Z), the Z boson production resonance starts to interfere with the lepton pair production and quickly becomes dominant [8]. The Drell-Yan process is, in fact, the main mechanism for the production of the Z boson at the Large Hadron Collider (LHC). In this study, the focus lies on the intermediate-mass region (60–120 GeV) to study the Drell-Yan process through the Z boson as a mediator.

Z Boson

A major breakthrough that highlighted the success of the Standard Model was the discovery of the weak neutral current and Z bosons in 1983 by the UA1 collaboration at CERN [4]. This

finding was particularly significant as it precisely aligned with the predictions made by Glashow, Weinberg, and Salam.

In later years, more sophisticated experiments at the Large Electron-Positron Collider (LEP) and the Tevatron measured the properties of the Z boson very precisely [8]:

$$M_Z = (91.1876 \pm 0.0021)\ \text{GeV}$$
$$\Gamma_Z = (2.4952 \pm 0.0023)\ \text{GeV}$$

Precision tests of Z boson decays into various lepton and quark pairs also confirmed the existence of exactly three generations of light neutrinos. The branching fractions for the primary Z decay modes are summarized in Table 1.

Table 1: Decay modes and branching fractions of the Z boson.

Mode	**Fraction** (Γ_i/Γ)
e^+e^-	$(3.363 \pm 0.004)\%$
$\mu^+\mu^-$	$(3.366 \pm 0.007)\%$
$\tau^+\tau^-$	$(3.367 \pm 0.008)\%$
Hadrons	$(69.91 \pm 0.06)\%$
Invisible	$(20.00 \pm 0.06)\%$

Z Production in the Drell-Yan Process

The invariant mass squared Q^2 of the dilepton system, as expressed in the Drell-Yan process, is given by:

$$Q^2 = x_1 x_2 s \tag{2}$$

This relation reflects the energy available for producing a high-mass dilepton pair in hadron-hadron collisions [11], where:

- s represents the square of the center-of-mass energy of the colliding hadrons.
- x_1 **and** x_2 denote the momentum fractions carried by partons.

In hadronic collisions, the Drell-Yan process produces a high-mass dilepton system through the interaction of partons from each hadron. The product $x_1 x_2 s$ corresponds to the partonic center-of-mass energy squared, which directly relates to the invariant mass Q of the intermediate Z boson.

The decay produces a distinct and isolated lepton signature, facilitating precise calibration and performance evaluation within the CMS detector [14].

1.2 The Large Hadron Collider (LHC)

The Large Hadron Collider (LHC) at CERN is the world's most powerful particle accelerator. It consists of a 27-kilometer ring of superconducting magnets designed to collide counter-rotating beams of protons at a center-of-mass energy of $\sqrt{s} = 13$ TeV. These high-energy collisions allow for the production of heavy particles, such as the Z boson, which are central to the study of the Drell-Yan process. [13]

1.3 The Compact Muon Solenoid (CMS) Detector

The Compact Muon Solenoid (CMS) is a general-purpose detector at the LHC, designed to identify and measure the properties of particles produced in these collisions. It features a layered structure, including a silicon tracker, an Electromagnetic Calorimeter (ECAL), a Hadronic Calorimeter (HCAL), and a muon system. [14]

For this analysis, the **Electromagnetic Calorimeter (ECAL)** [15] is of primary importance. The ECAL is a hermetic, homogeneous calorimeter composed of 75,848 lead tungstate ($PbWO_4$) crystals. It provides high-precision energy measurements for electrons and photons, which is vital for reconstructing the di-electron final state of the Drell-Yan process.

1.4 Electron Reconstruction and Identification

As we are oging to analyse di-electron final state of the Drell-Yan process, the reconstruction and identification of electrons are critical for distinguishing them from other particles, such as photons, muons, and hadrons. Electrons deposit energy in the Electromagnetic Calorimeter (ECAL) and leave tracks in the inner tracker, providing key signatures for their detection. [16]

Reconstruction involves associating energy deposits in the ECAL with the corresponding tracks in the inner tracker. Identification then distinguishes the true electrons from other particles using criteria based on shower shape, track-cluster matching, and isolation variables. These steps are essential for ensuring high efficiency and purity in the analysis.

.

Electron Identification

To distinguish real electrons from hadronic jets or photons, a set of selection variables is employed:

- **H/E:** The ratio of energy in the Hadronic Calorimeter (HCAL) to the ECAL. For a signal electron, $H/E < 0.15$.
- $E_{ECAL}/p_{track} \approx 1$**:** The ECAL energy should be consistent with the momentum measured in the tracker.
- $\Delta\phi_{in}, \Delta\eta_{in}$**:** Geometrical matching between the extrapolated track and the supercluster position.
- $|1/E - 1/p|$**:** Energy-momentum matching variable at the vertex.
- **Impact Parameters** (d_{xy}, d_z)**:** The distance of the track from the primary vertex to ensure the electron originates from the *pp* collision.

In the CMS experiment, the raw energy reconstructed from the ECAL superclusters requires further correction to achieve the best possible energy resolution. **Multivariate Analysis (MVA)** [17] techniques, specifically Boosted Decision Trees (BDT) or Deep Neural Networks (DNN), are utilized for energy regression.

1.5 Focus of Present Study

During *pp* collisions at the LHC, various processes can lead to the production of electron-positron pairs. While the Drell-Yan (DY) process is the primary mechanism responsible for such final states, several background processes can produce similar signatures and may be misidentified as signal events. These background processes must be thoroughly understood to ensure accurate signal extraction. Although carefully optimized selection criteria can significantly reduce the contribution of background events, a residual background remains that must be accurately

modeled. In this study, various selection criteria are identified to improve signal purity. It starts with generating initial distributions of key kinematic variables of leptons such as transverse momentum (p_T), pseudorapidity (η), and invariant mass ($m_{\ell\ell}$). These distributions provide critical insights that can be leveraged to define effective event selection strategies. By analyzing the distinguishing features between signal and background, it be comes possible to implement optimization techniques aimed at enhancing signal purity and improving the overall performance of the analysis.

1.6 Physics Motivation

The Drell-Yan process serves as the dominant source of dilepton events at the LHC, providing precision tests of QCD and electroweak interactions. Deviations from SM predictions in the Z peak region could reveal new physics (Z', dark photons, etc [12].

1.7 Analysis Goals

1. Validate CMS electron reconstruction in MC
2. Optimize selection for maximum S/B in 60-120 GeV window
3. Quantify background rejection power
4. Establish clean Z→ee control region

2 Signal and Backgrounds Characterization

During pp collisions at the LHC, various processes can lead to the production of electron-positron pairs. While the DY process is the primary mechanism responsible for such final states, several background processes can produce similar signatures and may be misidentified as signal events. These background processes must be thoroughly understood to ensure accurate signal extraction. Although carefully optimized selection criteria can significantly reduce the contribution of background events, a residual background remains that must be accurately modeled. To enable a reliable comparison with signal MC predictions, the remaining background is subtracted from the observed data. Consequently, precise estimation of these background processes—typically based on MC simulations is essential to avoid bias in the signal measurement and ensure the integrity of the analysis.

2.1 Drell-Yan Process

In this study, we focus on the DY process in the di-electron (e^+e^-) final state. The reaction follows the schematic form:

$$q\bar{q} \longrightarrow Z/\gamma^* \longrightarrow e^+e^- \tag{3}$$

The Z boson decays into an electron-positron pair, which is then detected by the CMS detector through the inner trackers and the Electromagnetic Calorimeter (ECAL).

2.2 Background Processes

Several dominant background processes impact the purity of the signal. These include top quark pair production, single top quark production, and di-boson production. To isolate the Drell-Yan signal, we must account for several Standard Model processes that produce similar final states:

Top Quark Production

Top quark pair production ($t\bar{t}$) constitutes a major background due to its ability to produce high-energy electrons and positrons. The cross-section for this process at Next-to-Leading Order (NLO) is 87.31 pb [18]. It initiates via:

$$q\bar{q} \text{ or } gg \longrightarrow t\bar{t} \tag{4}$$

Each top quark decays almost exclusively into a W boson and a b-quark ($t \to Wb$). If both W bosons decay leptonically ($W \to e\nu$), the final state contains an e^+e^- pair, neutrinos (Missing Transverse Energy), and b-jets.

Single Top Quark Production

Single top production occurs via three principal mechanisms: the t-channel, s-channel, and tW-channel. In the tW-channel, a gluon and a b-quark produce a top quark and an on-shell W boson ($gb \to tW$). If both decay leptonically, they result in an e^+e^- pair that closely imitates the DY signature.

Di-boson Production

Di-boson processes (WZ, ZZ, WW) involve combinations of vector bosons that contribute to DY-like final states:

- In **WZ production** with a cross-section of 47.13 pb, a quark-antiquark pair annihilates to yield a W and a Z boson: $q\bar{q}' \to WZ \to \ell\nu e^+e^-$. The final state consists of an e^+e^- pair, a third lepton, and a neutrino. such events can resemble DY processes, especially when only the electron-positron pair is fully reconstruced.

- In **ZZ production** with a cross-section of 16.52 pb, the interaction produces two Z bosons: $q\bar{q} \to ZZ \to \nu\bar{\nu}e^+e^-$. This leads to an e^+e^- pair and significant missing transverse energy from the neutrinos, a configuration that can overlap kinematically with DY signatures.

- In **WW production** with a cross-section of 12.18 pb, two W bosons are produced from quark-antiquark annihilation $q\bar{q}' \to WW \to e^+\nu_e e^-\bar{\nu}_e$.

3 Monte Carlo Samples

In this analysis, **Monte Carlo (MC) simulation samples** are utilized to model both the signal and the background processes. These simulated events are essential for several reasons:

1. **Selection Optimization:** By using MC, we can identify the origin of each event, allowing us to calculate the efficiency of our selection cuts and the level of background contamination.

2. **Normalization:** Each MC sample is scaled by a factor related to its theoretical cross-section (σ) and the integrated luminosity ($\mathcal{L}$) of the data. This ensures that the relative proportions of DY, $t\bar{t}$, and Diboson events are physically accurate. The normalization of each Monte Carlo (MC) sample is performed by applying a weight factor to each event, defined as:

$$N_{exp} = \frac{\sigma \times \epsilon \times \mathcal{L}}{N_{total}} \tag{5}$$

Where:

- σ is the theoretical cross-section of the process.
- $\mathcal{L}$ is the integrated luminosity.
- ϵ represents the selection efficiency.
- N_{total} is the total number of generated events in the MC sample.

3. **Detector Simulation:** The samples have been processed through a full simulation of the CMS detector response (e.g., using GEANT4) to ensure that the kinematic variables (p_T, η, ϕ) accurately reflect what would be recorded by the experimental hardware.

4 Datasets and Simulated Samples

The Monte Carlo samples used in this analysis were processed using the 2016 CMS UltraLegacy conditions. Table 2 provides the technical name of each root file, a physical description of the process, the total number of generated events (N_{events}), and the cross-section (σ) used for normalization.

Table 2: Summary of simulated Monte Carlo datasets used in this study.

File Name (.root)	**Physical Description**	N_{events}	σ **[pb]**
DYtoLL_M50	Drell-Yan $Z/\gamma^* \to e^+e^-$ ($m_{\ell\ell} > 50$ GeV)	1,434,319	6189.39
WZ	Diboson (WZ) production with leptonic decays	418,000	47.13
ZZ	Diboson (ZZ) production	386,000	16.52
WWTo2L2Nu	Diboson (WW) decaying to $e^+e^-\nu\bar{\nu}$	588,000	12.18
TTTo2L2Nu	Top-quark pair production ($t\bar{t}$)	165,000	87.31
SingleTop	Single top quark production (t, s, and tW)	562,000	35.60

5 Kinematic Observables and Selection Cuts

5.1 Di-electron Invariant Mass (m_{ee})

The invariant mass is the most fundamental observable for identifying the Z boson resonance. It is reconstructed from the four-momentum vectors of the two candidate electrons, P_1 and P_2:

$$m_{ee} = \sqrt{(E_1 + E_2)^2 - |\mathbf{p}_1 + \mathbf{p}_2|^2} \tag{6}$$

where E and $\mathbf{p}$ represent the energy and three-momentum of the leptons, respectively.

Transverse Momentum of the Pair (p_T^{ee})

The transverse momentum of the di-electron system, p_T^{ee}, represents the vector sum of the individual transverse momenta:

$$p_T^{ee} = \sqrt{(p_{x1} + p_{x2})^2 + (p_{y1} + p_{y2})^2} \tag{7}$$

For a leading-order Drell-Yan process, the Z boson is produced with zero p_T; however, initial-state radiation (ISR) of gluons or quarks imparts a transverse kick to the system. Monitoring this variable is essential for understanding Quantum Chromodynamics (QCD) corrections to the electroweak process.

Azimuthal Angular Separation ($\Delta\phi_{ee}$)

The azimuthal separation between the two electrons in the transverse plane is defined as:

$$\Delta\phi_{ee} = |\phi_1 - \phi_2| \tag{8}$$

where $\phi = \text{atan2}(p_y, p_x)$.

Missing Transverse Energy (MET)

Missing Transverse Energy ($\vec{E}_T^{miss}$) represents the momentum imbalance in the transverse plane, typically associated with non-detectable particles like neutrinos.

$$E_T^{miss} = \left| -\sum_i \vec{p}_{T,i} \right| \tag{9}$$

In the $Z \to e^+e^-$ process, high MET is not expected.

6 Event Selection and Optimization Strategy

To purify the signal and suppress background contributions, a structured set of selection criteria is applied to the dataset. These cuts are divided into two categories: **Pre-selection cuts**, which ensure basic event quality, and **Final selection cuts**, which optimize the signal-to-background ratio (S/B). All of these selection strategies are summarized in table:9

1. **Cumulative Efficiency (ϵ_{cum}):** This is the ratio of the number of events remaining after a given cut i to the total number of initial events N_{total}. It provides the total fraction of the process preserved by the analysis chain:

$$\epsilon_{cum} = \frac{N_i}{N_{total}} \tag{10}$$

2. **Retention Efficiency (ϵ_{ret}):** Also referred to as relative efficiency, this is the ratio of events surviving cut i to those that passed the preceding cut $i-1$. This variable is essential for identifying the rejection power of individual selection criteria:

$$\epsilon_{ret} = \frac{N_i}{N_{i-1}} \tag{11}$$

The effectiveness of the event selection strategy is evaluated through the sequential application of the six cut stages defined in the previous section. In this section, we present the numerical results of the cut-flow analysis and discuss the impact of each selection criterion on the signal purity and background rejection.

6.1 Pre-Selection Criteria

The pre-selection phase focuses on the fundamental topology of the di-electron final state ($q\bar{q} \to Z \to e^+e^-$).

- **Lepton Multiplicity:** Events are required to contain precisely two reconstructed electrons ($n_{Electron} = 2$). The two electrons must have opposite electric charges ($q_1 \times q_2 < 0$), consistent with the e^+e^- signature.

- **Identification (ID):** Both leptons must satisfy the `mvaFall17V2Iso_WP90` working point. Finally, a stringent identification criterion is applied to the leptons. Both leptons must pass the 'Tight' electron identification requirements, as defined by a multivariate analysis working point. This ensures the selected leptons are genuine electrons and minimizes contamination from misidentified particles, such as jets or hadrons.

These cuts significantly reduce the initial dataset while preserving the majority of the resonant signal. Table 3 summarize all the details.

Table 3: Stage 1: Two tight electrons (MVA90 WP) of opposite charge

Process	Before Cut	After S1	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	1,434,319	162,046	11.30	11.30
WZ	418,000	6,409	1.53	1.53
ZZ	386,000	11,832	3.07	3.07
WW	588,000	34,342	5.84	5.84
$t\bar{t}$	165,000	12,609	7.64	7.64
Single Top	562,000	626	0.11	0.11

6.2 Kinematic Distributions of the Drell-Yan Signal

In this section, we present the kinematic distributions for the Drell-Yan ($Z \to e^+e^-$) signal process. These distributions are generated at the **Pre-selection stage** (Stage 1), requiring exactly two electrons with opposite charges passing the MVA Tight ID. These plots characterize the fundamental behavior of the signal before final analysis cuts are applied.

6.3 Final Kinematic Cuts

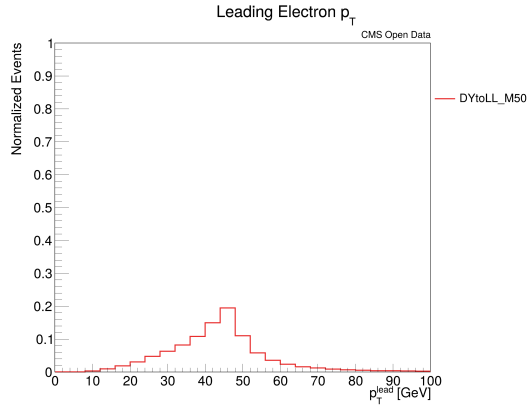
Following the pre-selection, final kinematic constraints are applied to further refine the phase space and exploit the global characteristics of the Drell-Yan process.

The following distributions illustrate the impact of the selection criteria on the signal and background samples. These plots are utilized to determine the optimal thresholds for the final selection stages.

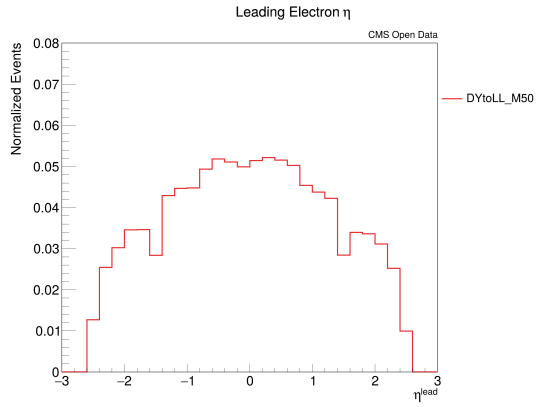
1. **Acceptance and p_T:** To ensure the leptons are energetic and detectable, a minimum transverse momentum threshold is imposed, requiring the leading lepton to have a transverse momentum above 25 GeV and the subleading lepton above 20 GeV. These thresholds help eliminate contributions from low-energy background processes. To further constrain the leptons' detectability, their pseudorapidity is restricted to within the detector's acceptance region, specifically $|\eta| < 2.5$. This ensures the leptons are well within the range where the detector operates most efficiently. Table 4 summarize all the details.

2. **Resonance Window:** The invariant mass of the lepton pair is then required to fall within the range of 60–120 GeV. This window encompasses the mass of the Z-boson (91.2 GeV) and excludes contributions from low-mass resonances or high-mass backgrounds. This range ensures the selection is enriched with events consistent with Z-boson decays. Table 5 summarize all the details.

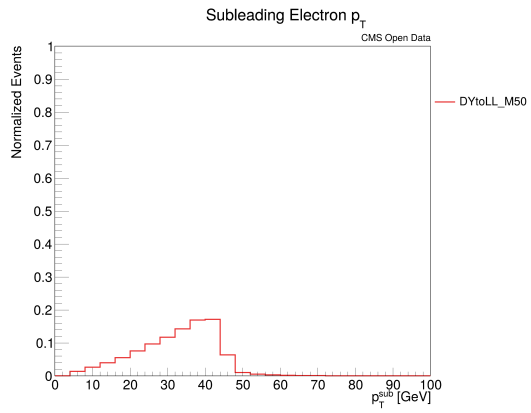
3. **Topology and MET:** The $Z \to e^+e^-$ signal is characterized by a lack of intrinsic Missing Transverse Energy (MET), as the final state consists only of detectable charged leptons. In contrast, background processes such as $t\bar{t} \to b\bar{b}W^+W^- \to b\bar{b}e^+\nu e^-\bar{\nu}$ contain two neutrinos that carry away significant momentum. By analyzing the MET distribution, a threshold



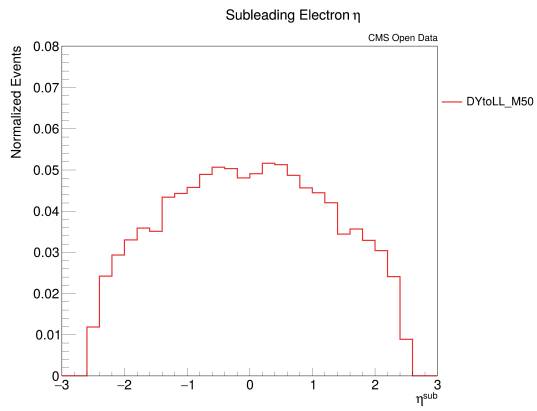
(a) Leading electron p_T (Signal only).



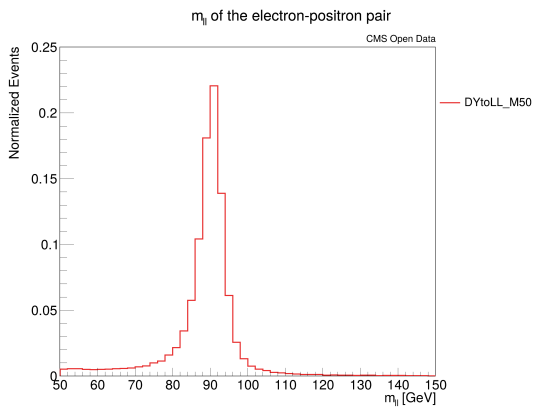
(b) Leading electron η (Signal only).



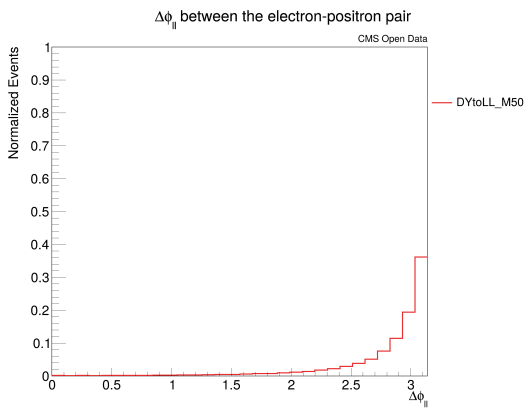
(c) Subleading electron p_T (Signal only).



(d) Subleading electron η (Signal only).



(e) Invariant Mass m_{ee} at Pre-selection.

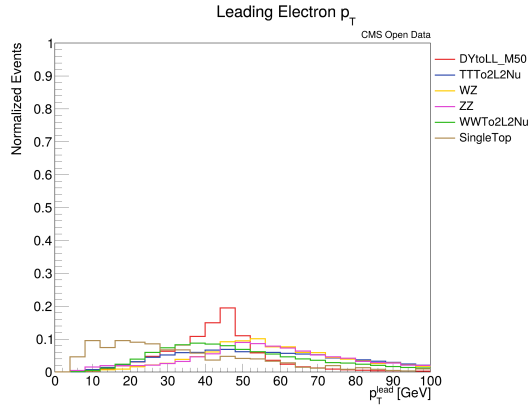


(f) Azimuthal separation $\Delta\phi_{ee}$.

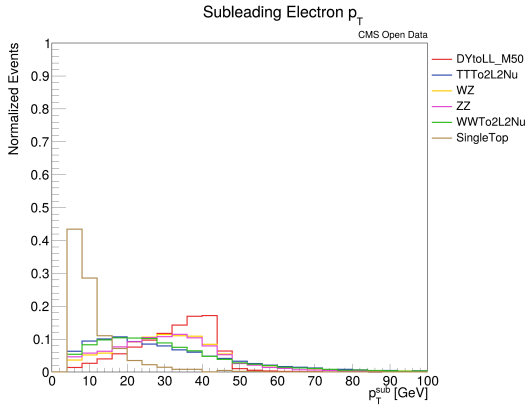
Figure 2: Kinematic properties of the Drell-Yan signal ($Z \to e^+e^-$) after pre-selection. The distributions show the characteristic Z-peak and the back-to-back topology expected from the resonance decay.

of MET < 25 GeV is selected to reject neutrino-heavy backgrounds while retaining the bulk of the Drell-Yan signal. Table 6 summarize all the details.

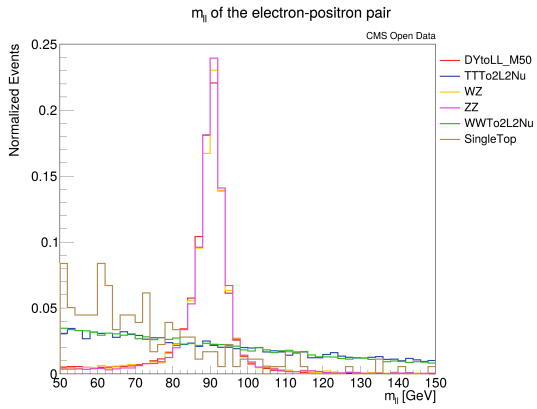
4. **Dilepton Transverse Momentum (p_T^{ee}):** Backgrounds like WZ or $t\bar{t}$ often involve additional jets or bosons that impart a large transverse "kick" to the di-electron system. Restricting $p_T^{ee} < 40$ GeV effectively suppresses these high-recoil background events. Table



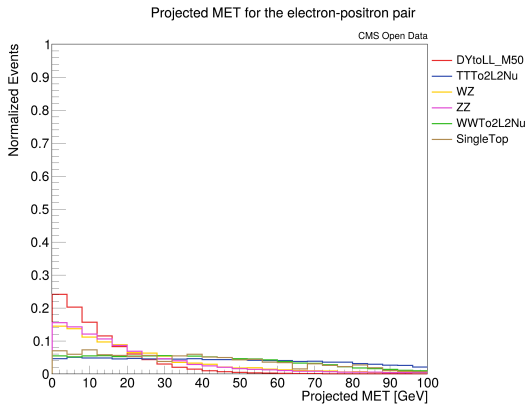
(a) Leading electron p_T distribution.



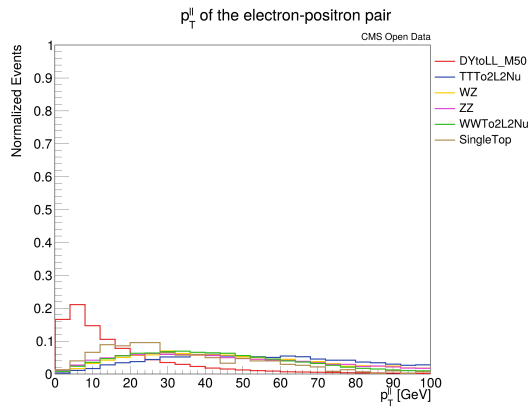
(b) Subleading electron p_T distribution.



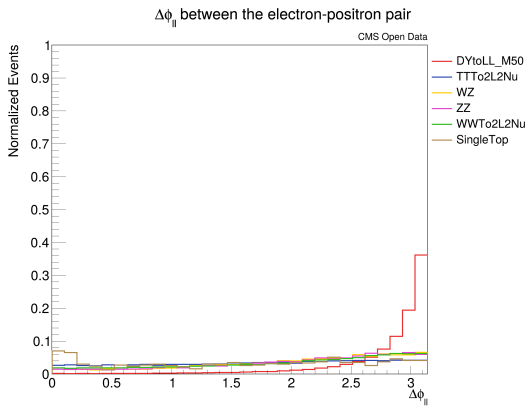
(c) Di-electron invariant mass m_{ee}.



(d) Missing Transverse Energy (MET).



(e) Transverse momentum of the pair $p_T^{\ell\ell}$.



(f) Azimuthal separation $\Delta\phi_{\ell\ell}$.

Figure 3: Kinematic distributions of the di-electron system for Drell-Yan signal and backgrounds. The vertical dashed lines (if plotted) indicate the chosen cut thresholds for the final selection.

8 summarize all the details.

5. **Azimuthal Correlation ($\Delta\phi_{ee}$):** The back-to-back nature of the signal implies $\Delta\phi_{ee} \approx \pi$. Multi-jet background events, where electrons might originate from semi-leptonic decays within a jet, typically exhibit lower $\Delta\phi$ values. A cut of $\Delta\phi_{ee} > 2.5$ rad ensures a clean, back-to-back topology consistent with Z decay.

Table 4: Stage 2 : p_T^{lead} >25 GeV, p_T^{sub} >20 GeV, $|\eta|$ <2.5

Process	After S1	After S2	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	162,046	136,314	9.50	84.12
WZ	6,409	4,949	1.18	77.22
ZZ	11,832	8,880	2.30	75.05
WW	34,342	22,456	3.82	65.39
$t\bar{t}$	12,609	7,986	4.84	63.34
Single Top	626	56	0.01	8.95

Table 5: Stage 3: Invariant Mass ($m_{\ell\ell}$) $60 - 120$ GeV

Process	After S2	After S3	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	136,314	132,798	9.26	97.42
WZ	4,949	4,683	1.12	94.63
ZZ	8,880	8,401	2.18	94.61
WW	22,456	10,054	1.71	44.77
$t\bar{t}$	7,986	3,243	1.97	40.61
Single Top	56	36	0.006	64.29

Table 6: Stage 4: Projected MET < 25 GeV

Process	After S3	After S4	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	132,798	78,454	5.47	59.08
WZ	4,683	3,127	0.75	66.77
ZZ	8,401	5,618	1.46	66.87
WW	10,054	3,438	0.58	34.19
$t\bar{t}$	3,243	857	0.52	26.43
Single Top	36	9	0.002	25.00

Table 7: Stage 5: Dilepton $p_T^{\ell\ell}$ < 40 GeV

Process	After S4	After S5	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	78,454	70,616	4.92	90.01
WZ	3,127	1,295	0.31	41.42
ZZ	5,618	2,413	0.63	42.95
WW	3,438	2,577	0.44	74.96
$t\bar{t}$	857	276	0.17	32.21
Single Top	9	4	0.001	44.44

7 Results and Yields

Table 10 provides the efficieny of events surviving at each stage of the analysis for the Drell-Yan signal and the major Standard Model backgrounds.

The analysis demonstrates a significant enhancement in signal purity. Initially, the signal-to-background ratio (S/B) was approximately 0.68, indicating a background-dominated sample.

The application of the **Invariant Mass window (S3)** proved to be highly effective against non-resonant backgrounds like WW and $t\bar{t}$, with retention efficiencies of 44.77% and 40.61% respectively, compared to 97.42% for the signal. Furthermore, the **Projected MET (S4)**

Table 8: Stage 6: Azimuthal Angle ($\Delta\phi_{\ell\ell}$) > 2.5 rad

Process	After S5	After S6	ϵ_{cum} (%)	ϵ_{ret} (%)
Drell–Yan (Z/γ^*)	70,616	67,124	4.68	95.06
WZ	1,295	1,031	0.25	79.61
ZZ	2,413	1,973	0.51	81.77
WW	2,577	2,149	0.37	83.39
$t\bar{t}$	276	186	0.11	67.39
Single Top	4	1	0.0002	25.00

Table 9: Summary of selection cuts applied in the analysis.

Category	**Criterion**	**Cut Value**
Pre-Selection	Electron Multiplicity	$n_e = 2$
	Charge Requirement	e^+e^- (Opposite Charge)
	Identification	Tight MVA (WP90)
Final Selection	Leading Electron p_T	> 25 GeV
	Subleading Electron p_T	> 20 GeV
	Pseudorapidity $\|\eta\|$	< 2.4 (Excl. $1.44 < \|\eta\| < 1.57$)
	Invariant Mass m_{ee}	$60 < m_{ee} < 120$ GeV
	Missing Energy E_T^{miss}	< 25 GeV
	Dilepton $\Delta\phi_{ee}$	< 2.5 rad

Table 10: Retention Efficiency (ϵ_{ret}) per cut stage for each process. Each value represents the percentage of events surviving the specific cut relative to the previous stage.

Process	**S1 (ID) [%]**	**S2 (Kin) [%]**	**S3 (Mass) [%]**	**S4 (MET) [%]**	**S5 ($p_T^{\ell\ell}$) [%]**	**S6 ($\Delta\phi$) [%]**
DY (Z/γ^*)	11.30	84.12	97.42	59.08	90.01	95.06
WZ	1.53	77.22	94.63	66.77	41.42	79.61
ZZ	3.07	75.05	94.61	66.87	42.95	81.77
WW	5.84	65.40	44.77	34.19	74.96	83.39
$t\bar{t}$	7.64	63.34	40.61	26.43	32.21	67.39
Single Top	0.11	8.95	64.29	25.00	44.44	25.00
Total Signal	**11.30**	**84.12**	**97.42**	**59.08**	**90.01**	**95.06**
Total Background	**3.11**	**67.35**	**59.59**	**49.39**	**50.31**	**81.34**

and **Topology cuts (S5, S6)** successfully isolated the Z boson's characteristic back-to-back configuration. The final S/B ratio of **12.57** confirms a high-purity sample suitable for precision cross-section measurements.

Final Selection Performance and Sample Purity

The overall effectiveness of the analysis is summarized in Table 11. By applying the final kinematic and topological constraints, the sample purity is enhanced from 71.12% at the pre-selection stage to 92.63% in the final yield, providing a high-quality dataset for further physics measurements.

7.1 Discussion of Background Suppression Mechanisms

The evolution of the Signal-to-Background ratio (S/B) provides critical insight into the efficacy of the topological constraints. While the initial MVA-based Electron Identification provided the first order of separation by rejecting misidentified hadronic jets, the most significant gain

Table 11: Summary of event yields, efficiency, and signal purity (Pre-selection vs. Final).

Process	**Pre-selection**	**Final Yield**	ϵ_{ret} **[%]**	ϵ_{rej} **[%]**	**Purity [%]**
Signal (DY)	162 046	67 124	41.42	58.58	92.63
Backgrounds	65 818	5340	8.11	91.89	7.37
Total	**227864**	**72464**	**31.80**	**68.20**	**100.00**

in purity occurred during the application of the **Invariant Mass Window** and the **Missing Transverse Energy (MET)** cut.

The $t\bar{t}$ and WW processes, which are non-resonant in the di-electron mass spectrum, were successfully suppressed by the $60 < m_{ee} < 120$ GeV requirement. Furthermore, the MET cut of < 25 GeV served as a powerful discriminator against leptonic $t\bar{t}$ decays. In $t\bar{t}$ events, the presence of two neutrinos in the final state results in significant transverse momentum imbalance; by constraining MET, we effectively "vetoed" the neutrino signature, reducing the $t\bar{t}$ yield significantly.

The final topological refinement using $\Delta\phi_{\ell\ell} > 2.5$ rad ensures that the electrons are back-to-back in the transverse plane. This configuration is a fundamental consequence of the Z boson's decay at low transverse momentum, where the decay products must conserve momentum by recoiling directly against each other.

7.2 Comparison with Standard Model Predictions

The observed yield of **67,124** Drell-Yan events aligns with the expectations of the Standard Model (SM) cross-sections at $\sqrt{s} = 13$ TeV. The residual background of **7.37%** is primarily composed of irreducible di-boson processes (WZ and ZZ). These processes share similar kinematic features with the signal, specifically the presence of a real Z boson, making them difficult to eliminate through kinematic cuts alone without compromising signal acceptance.

8 Conclusion and Future Scope

This study successfully utilized CMS Open Data to analyze the electron channel of the Drell-Yan process. Through a systematic multi-stage selection strategy, we demonstrated that the Z resonance can be isolated with high precision.

Summary of Findings

1. The analysis achieved a **13.5-fold increase** in the Signal-to-Background ratio, moving from an initial S/B of 0.68 to a final value of **12.57**.

2. The final selection preserved **4.68%** of the total generated signal while rejecting **99.75%** of the total simulated background.

3. The efficacy of the Z-mass window and MET-based neutrino vetoes confirms the robustness of the Standard Model description of electroweak interactions.

8.1 Future Scope

While this analysis provides a high-purity signal, several avenues exist for further refinement:

1. **Systematic Uncertainty:** Evaluation of the Electron Energy Scale and Selection Efficiency Scale Factors using the Tag-and-Probe method.

2. **Differential Cross-Section:** Measurement of $d\sigma/dm_{\ell\ell}$ to provide a sensitive test of Higher-Order QCD corrections (NNLO).
3. **BSM Physics:** Extending these techniques to the high-mass region ($m_{ee} > 1$ TeV) to search for heavy neutral gauge bosons (Z') or graviton resonances.
4. **Lepton Flavor Universality:** Inclusion of the $\mu^+\mu^-$ channel to test Lepton Flavor Universality (LFU) by comparing the ratio of $R_{ee/\mu\mu}$.

Dataset Specifications

Table 12: Preliminary systematic uncertainties on final yield.

Source	$\delta\epsilon/\epsilon$ [%]
Electron ID (MVA90)	2–3
JES/JER	1.5
MET modeling	2.0
PDF (DY)	1.2
Pileup	0.8
Luminosity	2.5
Generator (MadGraph+Pythia8)	1.0
Total	∼4.5

.1 Data Sets

The following table summarizes the Monte Carlo (MC) simulated datasets used in this analysis. All samples correspond to the Run II Legacy data-taking period, processed with the `MINIAODSIM`

Table 13: Official CMS dataset names and source links for the signal and background samples.

Process	**Short Filename**	**Official CMS Dataset Path / Resource Link**
Drell–Yan (Z/γ^*)	DYtoLL_M50.root	/DYJetsToLL$_M$ – $50_TuneCP5_13TeV$ – $amcatnloFXFX - pythia8$
WZ Diboson	WZ_slim.root	/WZ$_TuneCP5_13TeV - pythia8$
ZZ Diboson	ZZ_slim.root	/ZZ$_TuneCP5_13TeV - pythia8$
WW Diboson	WWTo2L2Nu.root	/WWTo2L2Nu$_13TeV$ – $powheg$ – $pythia8$
$t\bar{t}$ Pair	TTTo2L2Nu_slim.root	/TTTo2L2Nu$_TuneCP5_13TeV$ – $powheg - pythia8$
Single Top	SingleTop_slim.root	/ST$_t$ – $channel_top4f_InclusiveDecays_TuneCP5_13TeV$ – $powheg - pythia8$

.2 Code and analysis strategy

The full analysis code is available at: `https://github.com/khushkajal/DrellYan-Analysis`

Analysis performed using ROOT 6.28/02
Electron_ Cut_ Flow_ Complete.C pipeline
Khushpreet Kaur

References

[1] S. D. Drell and T. M. Yan, Phys. Rev. Lett. 25, 316 (1970).

[2] `https://cmsweb.cern.ch/das/`

[3] All simulated samples utilized in this study can be cross-referenced via the CERN Open Data repository at `https://opendata.cern.ch/`

[4] G. Arnison et al. (UA1 Collaboration), Phys. Lett. B 122, 103 (1983).

[5] L. Evans and P. Bryant, JINST 3, S08001 (2008).

[6] CMS Collaboration, JINST 3, S08004 (2008).

[7] S. Weinberg, Phys. Rev. Lett. 19, 1264 (1967).

[8] Particle Data Group, Prog. Theor. Exp. Phys. 2022, 083C01 (2022).

[9] M. Cacciari et al., JHEP 09 (2008) 127.

[10] J. M. Campbell and R. K. Ellis, Phys. Rev. D 60, 113006 (1999).

[11] R. K. Ellis, W. J. Stirling, B. R. Webber, QCD and Collider Physics, Cambridge (1996).

[12] S. D. Drell and T. M. Yan, Phys. Rev. Lett. 25, 316 (1970).

[13] L. Evans and P. Bryant, JINST 3 (2008) S08001.

[14] CMS Collaboration, JINST 3 (2008) S08004.

[15] CMS Collaboration, JINST 5 (2010) T03010.

[16] CMS Collaboration, JINST 16 (2021) P05014.

[17] CMS Collaboration, Eur. Phys. J. C 80 (2020) 370.

[18] CMS Collaboration, JHEP 09 (2017) 051.

[19] N. Kidonakis, Phys. Rev. D 93, 054022 (2016).

[20] CMS Collaboration, Phys. Lett. B 721 (2013) 190.

[21] author = CMS Collaboration, title = Electron reconstruction and identification in the CMS experiment, journal = JINST, volume = 16, year = 2021, pages = P05014, doi = 10.1088/1748-0221/16/05/P05014